Second Examination 2025
Artificial Intelligence and Machine Learning

1. Given a data set contained in the file named "data.csv", please apply Principal Component Analysis (PCA) to project the 25-dimensional data (with 1 additional data showing associated classes) into a 2-dimensional space and plot the dimensionality reduction results separately. In your plot, please use the first major eigenvector of $\boldsymbol{\Sigma}$ as the X-axis and the second major eigenvector of $\boldsymbol{\Sigma}$ as the Y-axis. Please write a report with the file name "report1.pdf" showing (a) the process details you did for this problem, (b) the largest 2 eigenvalues of $\boldsymbol{\Sigma}$, (c) the projection matrix $\boldsymbol{W}$, (d) the plot of all data projected onto the 2-dimensional space. Please use dots with different colors to show the data of different classes, and provide legend explaining the meaning of colors. In addition to the file "report1.pdf", please also provide the pdf file named "source1.pdf" containing your source code. Note that the plot should be put into the file "report1.pdf" along with the process details. (15%)
2. Given a data set contained in the file named "data.csv", please apply Linear Discriminant Analysis (LDA) to project the 25-dimensional data (with 1 additional data showing associated classes) into a 2-dimensional space and plot the dimensionality reduction results separately. In your plot, please use the major eigenvector of $\boldsymbol{S_W^{-1}S_B}$ as the X-axis and the second major eigenvector of $\boldsymbol{S_W^{-1}S_B}$ as the Y-axis. Please write a report with the file name "report2.pdf" showing (a) the process details you did for this problem, (b) the largest 2 eigenvalues of $\boldsymbol{S_W^{-1}S_B}$, (c) the projection matrix $\boldsymbol{W}$, (d) the plot of all data projected onto the 2-dimensional space. Please use dots with different colors to show the data of different classes, and provide legend explaining the meaning of colors. In addition to the file "report2.pdf", please also provide the pdf file named "source2.pdf" containing your source code. Note that the plot should be put into the file "report2.pdf" along with the process details. (15%)
3. Given a data set contained in the file named "smpldata.csv", please apply Laplacian Eigenmaps (LE) to project the 18-dimensional data (with 1 additional data showing associated classes) into a 2-dimensional space and plot the dimensionality reduction results separately. Note that the standard deviation σ in the Gaussian kernel converting Euclidean distance to a similarity value is set to be 0.005. In your plot, please use the eigenvector corresponding to the smallest non-zero eigenvalue of $\boldsymbol{L}$ as the X-axis and the eigenvector corresponding to the second smallest non-zero

eigenvalue of $\boldsymbol{L}$ as the Y-axis. Please write a report with the file name "report3.pdf" showing (a) the process details you did for this problem, (b) the 2 smallest non-zero eigenvalues of $\boldsymbol{L}$, (c) the projection matrix $\boldsymbol{W}$, (d) the plot of all data projected onto the 2-dimensional space. Please use dots with different colors to show all data, and provide legend explaining the meaning of colors. In addition to the file "report3.pdf", please also hand in the pdf file named "source3.pdf" containing your source code. Note that the plot should be put into the file "report3.pdf" along with the process details. (15%)

4. Given a data set contained in the file named "smpldata.csv", please apply Multi-dimensional Scaling (MDS) to project the 18-dimensional data (with 1 additional data showing associated classes) into a 2-dimensional space and plot the dimensionality reduction results separately. In your plot, please use the major eigenvector of $\boldsymbol{YY^T}$ as the X-axis and the second major eigenvector of $\boldsymbol{YY^T}$ as the Y-axis. Please write a report with the file name "report4.pdf" showing (a) the process details you did for this problem, (b) the largest 2 eigenvalues of $\boldsymbol{YY^T}$, (c) the plot of all data projected onto the 2-dimensional space. Please use dots with different colors to show all data, and provide legend to explaining the meaning of colors. In addition to the file "report4.pdf", please also provide the pdf file named "source4.pdf" containing your source code. Note that the plot should be put into the file "report4.pdf" along with the process details. (15%)
5. Given a data set contained in the file named "clstr_data.csv", please apply k-means to find the cluster centers. Assume that there are 7 clusters in the data. The initial cluster centers are all located at the **origin**. After learning your clustering algorithm, please test your clustering algorithm on the data named "clstr_tst.csv" and put the clustering results in the last column of the testing data. In order to assign a cluster number for every cluster, 14 sample data at the beginning of the training data file "clstr_data.csv" have been assigned a cluster data, respectively. You may name the data with the same cluster number if the data belong to the same cluster as those 14 sample data. Please write a report with the file name "report5.pdf" showing (a) the process details you did; (b) the convergent coordinates of these 7 cluster centers; (c) explanation of the criteria you made in your source code in order to obtain the convergent solutions. (d) Please generate a file named "result5.csv" to contain the testing results based on the testing data in the file "clstr_tst.csv". Each row of the data in the file "result5.csv" should contain [testing data x], [cluster r]. In addition to the file "report5.pdf", please also hand in the pdf file named "source5.pdf" containing your source code. (15%)
6. Given a dataset contained in the file named "EM_data.csv", please apply Expectation Maximization method to find the cluster center and covariance matrix

of every cluster. Test your EM on the testing dataset in the file named "EM_tst.csv". Assume that there are 7 clusters in the data. Please use K-means to determine the initial conditions for EM. You may start your K-means with all initial cluster centers locating at origins, i.e. **0** vectors. After learning your EM algorithm, please test your EM algorithm on the data named "em_tst.csv" and put the clustering results in the last column of the testing data. In order to assign a cluster number for every cluster, 14 sample data at the beginning of the training data file "em_data.csv" have been assigned a cluster data, respectively. You may name the data with the same cluster number if the data belong to the same cluster as those 14 sample data. Please write a report with the file name "report6.pdf" showing (a) the process details you did; (b) the initial conditions of EM that have been obtained using K-means. (c) the convergent cluster center $\boldsymbol{m}_i$ and covariance matrix $\boldsymbol{S}_i$ of every i-th cluster, i = 1…7; (d) explanation of the criteria you made in your source code in order to obtain the convergent solutions. (e) Generate a file named "result6.csv" containing the testing results based on the EM you learned. Each row of the data in the file "result6.csv" should contain [testing data x], [cluster r]. In addition to the file "report6.pdf", please also hand in the pdf file named "source6.pdf" containing your source code. (25%)

You should upload all files to your account in Google Drive no later than 22:00 pm, Nov. 29, 2025. Any files submitted later than the due time will not be accepted.